1-7 Review of Mathematical Theory

Theory of Computation

Let us discuss each one by one -

i) Negation

horizontal bar on letter.

If S is a proposition then S defines NOT S. The truth table for this connective is

S	1s
T	F
F	T

ii Conjunction AND

If A and B are two propositions then conjunction of A and B is denoted by A^B.The

truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F

A	B	A^B
F	F	F

iii) Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB.The

truth table for this cornective is

A	B	AvB
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted

byA-BorA=B.

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Theory of Computation

The truth table for this connective is AB

A
B
A
T
T
F
T
T
F
F
F

T

F

T

T

For example: Find truth value for following proposition.

If an apple is not green then it must be red".

Let, A denotes "An apple is not green" and B denotes "It must be red". The statement

A is false and B is true. Hence the proposition is true.

v)If and only if

If A and B are two propositions then A if and only if B. It can be denoted as A B

The truth table will be

Col_1	B	AB
A		
	T	T
T		
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not raining.Here statement A

is I will go to the school. The statement B is It is not raining Thus the resultant

proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects even this is true in our real life.

One olfect can be related with the other object by a some relation.Then those two objects

forma pair based on this certain relationship.

Deinition The relation R is a collction for the set S which represents the pair of

elements.

For example: (a b) is in R. We can represent their relation as a R b.The first

pair is chosen from a set called range.

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Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk.
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive

For exampleif A=ab| then

reflexive relation R can be=a, a)b, b

irreflexive relation R can be =|a b)1

transitive relation R can be=ab.b, a)a, a1

symmetric relation R can be=ab)b, a

asymmetric relation R can be=ab|

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive symmetric and transitive,

A relaton 15

over some set S.

Suppose R is a set of relations and S is the

A relaton 15

set of elements.

For example :S is the set of lines in a plane and R is the relation of lines intersecting

to each other

A relation is said to be antisymmetric if a,b e R and b,a)e R then a =a

Example 1.5: Determine whether R is equivalence relation or not where

A=0121R=0010)11)2,221

SolutionThe R is reflexive because 0,0)1,1).2,2e R.

Where as R is not symmetric because 0,1) is not in R whereas (1, 0) is in R

Hence the R is not a equivalence relation

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive.

For exampleLeta,b)bcaa),b,b